MIL-D-3358C(ME)
20 December 1977
SUPERSEDING
MIL-D-3358B(ME)
2 April 1971

MILITARY SPECIFICATION

DRILLING MACHINES, RAIL, GASOLINE OR DIESEL, PORTABLE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a one-man-operated, portable, gasoline- or diesel-engine-driven, rail drilling machine.

1.2 Classification.

Class I - Gasoline engine driven.
Class II - Diesel engine driven.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 2230

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment.)
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Aluminum and Aluminum Alloys.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-M-18058	- Machinery, Metal and Wood-Working, Support Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.

STANDARDS

MILITARY

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MS15003	- Fittings, Lubrication (Hydraulic) Surface Check, 1/8 Pipe Threads, Steel, Type III.
MS24207	- Coupling, Grease Gun; Hydraulic Type Nozzle.
MS35647	- Padlocks, Key Operated.
MS35844	- Plug, Machine, Thread, Magnetic Drain.
MS49006	- Plugs, Pipe, Magnetic - Headless; Aluminum, Iron or Steel.

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A1 - Open Hearth Steel Rails.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN WELDING SOCIETY, INC.

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 NW 7th Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 45th Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

3. REQUIREMENTS

3.1 Description. The drilling machine shall be a one-man-operated, portable, gasoline-or diesel-engine-driven, rotary drill, mounted on a frame with rollers. The machine shall operate on 36-, 39-3/8-, 42-, 56-, 60-, 63-, and 66-inch (914-, 994-, 1067-, 1435-, 1524-, 1600-, and 1676-mm) gage tracks.

3.2 First article.

3.2.1 Preproduction model. The contractor shall furnish a drilling machine for examination and test within the time frame specified to prove, prior to starting production, that his production methods and choice of design detail will produce drilling machines that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4).

3.3 Initial production. When specified, the contractor shall furnish to the Government one or more drilling machines for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Rubber products furnished shall not exceed 8 calendar quarters in age from the date of cure of the rubber to the date of acceptance of the drilling machine. Unless protected against electrolytic corrosion, dissimilar metal material, as defined in MIL-STD-889, shall not be used in intimate contact with each other. Threaded parts shall

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be in accordance with SAE standards and Handbook H28. Maximum practicable use shall be made of interchangeable hardware and fastening devices and of a minimum number of types and sizes of studs, bolts, screws, washers, nuts, sumps, and similar openings on the drilling machine shall prevent the entrance of airborne water and airborne dust (see 6.5).

3.5 Environmental and use requirements. The environmental and use requirements shall be as specified herein. The criteria for design, environment, application, and use shall be as specified herein.

3.5.1 Operating temperature. Unless otherwise specified herein the drilling machine shall start and operate in any ambient temperature from minus 25° F to plus 120° F (minus 31.7° C to plus 48.9° C).

3.6 Safety. The safety characteristics of the drilling machine shall meet the following minimum requirements:

(a) All belts, gears, shafts, pulleys, sprockets, spindles, flywheels, chains, and other reciprocating, rotating, or moving parts of equipment shall be guarded when such parts are exposed to contact by personnel or otherwise create a safety hazard.
(b) All hot surfaces of equipment, including exhaust pipes and other lines which may be subjected to high temperatures, exposed to contact by personnel or which create a fire hazard shall be fully guarded or insulated.
(c) Exhaust or discharges from equipment shall be directed away from operating personnel.
(d) Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided.

3.7 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments. Covers, plates, or panels which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with clinch- or weld-nuts, threaded inserts, quick-disconnect fasteners, or threaded base members. Removal of all fasteners shall be accomplished by one mechanic without requiring the assistance of a helper. Where incorrect installation of an item could cause malfunction of that item or of the system in which it is installed, an unsymmetrical mounting means shall be provided or the mating parts shall be doweled or matchmarked.

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3.8 Overall dimensions and weight. The overall dimensions, not including stabilizing bar when furnished, and weight of the drilling machine shall not exceed the following:

Length	46-inches (1168 mm)
Height	30-inches (762 mm)
Width	24-inches (610 mm)
Weight	175-pounds (79 kg)

3.9 Operating characteristics.

3.9.1 Functional requirements. The drilling machine shall mount on and be able to drill rails weighing from 80 to 155 pounds (36 to 70 kg) linear yard (914 mm).

3.9.2 Drilling speed. The drilling machine shall drill a hole through the web of a 115 pounds-(52 kg)-per-yard (914 mm) rail conforming to ASTM A1, with a 1-1/8 inch (29 mm) diameter drill bit as specified in 3.11 and in not more than 1-1/2 minutes.

3.9.3 Drilling accuracy. The drilling machine shall drill holes perpendicular to a vertical plane running the length of a rail within plus or minus 1/2 degree. Hole diameter shall be accurate to 1/64-inch (1 mm).

a rail, shall move the drill up to the rail by manual or automatic means and shall control the feed during drilling. When the feed is automatic, a hand feed which shall override the automatic feed shall also be furnished.

3.9.5 Operation. The drilling machine shall be capable of being installed and operated on 80 to 155 pounds-(36 to 70 kg)-per-yard (914 mm) single rails and near switches where the two rails are 14-inches (356 mm) apart. Removal of the drilling machine when positioned near switches where the rails are 14-inches (356 mm) apart shall be accomplished by one man in not more than 10 seconds.

3.9.6 Capacity. The drilling machine shall operate as specified herein when drilling with 3/4-, 7/8-, 1-, and 1-1/8-inch (19-, 22-, 25-, 29 mm) drill bits.

3.10 Frame. The frame shall support the drilling machine when installed for drilling operations. The frame shall be of cast, forged, or fabricated material machined to keep the drilling machine components in alinement.

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3.11 Drill bits. Drill bits shall be of the flat-beaded type for their entire length. Overall length of the drill bits shall be 156-mm (6-1/4-inch) plus or minus 1/32-inch (1 mm). Two drills of each size 3/4-, 7/8-, 1-, and 1-1/8-inch (19-, 22-, 25-, and 29 mm) shall be furnished.

3.12 Drill chuck. The drill chuck shall be of the self-tightening type and shall hold the flat-beaded drills specified in 3.11. The bits shall be installed and removed without the use of tool. The chuck shall permit the use of bits which have an overall length of 6-7/32 to 6-9/32 inches (158 to 160 mm).

3.13 Engine. For the purpose of selecting the engine, the drilling machine shall be classified as a Class I end item in accordance with MIL-STD-1410. The engine shall be either gasoline or diesel as specified and shall conform to MIL-STD-1410, Class I. A fuel tank shall be furnished with a capacity for not less than 8 hours continuous operation at working speed. When a diesel engine is specified, it shall include an electrical cranking system and oil pressure and ammeter indicators.

3.14 Power transmission. The drilling machine shall be equipped with either a self-energizing friction type clutch or a hydrodynamic drive unit.

3.14.1 Clutch. A clutch of the self-energizing, friction type shall be provided. The clutch shall be located between the engine and the power-transmission housing. The clutch capacity shall transmit not less than 150 percent of the maximum torque developed by the engine.

3.14.2 Hydrodynamic drive unit. The hydrodynamic drive unit (torque converter or fluid coupling) shall be of commercial design and shall be matched to the power and speed of the engine furnished as specified herein. The hydrodynamic drive unit and its installation shall be in accordance with the drive unit manufacturer's recommendations. The hydrodynamic drive unit shall be located within the power train such that, when engine speed is at the maximum governed rpm the speed of the hydrodynamic drive unit shall be not less than 80 percent of the engine speed with the flexible shaft unloaded. When specified, a hydrodynamic drive unit only shall be supplied.

3.15 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.15.1 Lubricants. The drilling machine shall operate as specified herein when lubricated with military lubricants (see 6.3).

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3.15.2 Lubrication fittings. Lubrication fittings shall conform to MS15003. Fittings shall be located in a protected position. Fittings shall be accessible to a grease gun conforming to MIL-G-3859, equipped with a hydraulic coupler conforming to MS24207 and a flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Engine housing side panels and plates equipped with hand-operable quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.15.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.15.4 Enclosures with integral reservoir of lubricant. Enclosures such as gearcases and transmission housings which contain a reservoir of lubricants for lubrication of the parts enclosed shall be equipped with dipsticks, finger holes, or sight holes to determine the level of lubricant. Each enclosure shall be equipped with a means for filling the enclosure with lubricant and for draining. The drain shall be fitted with a drain plug conforming to MS35844 or MS49006. In lieu of a magnetic drain plug, a separate magnetic device shall be fastened inside the enclosure at or near the bottom and shall be removable for cleaning without removal of other parts or components. The drain plug shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the drilling machine is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain to the ground. Accessibility to the drain plug, the filling means, and the lubricant level checking device shall be provided.

3.15.5 Lubrication service.

3.15.5.1 Grease lubrication. Grease bearing lubrication, including lubrication of sealed bearings, shall be with greases conforming to MIL-G-23827 or MIL-G-10924 as applicable. The drilling machine shall be run-in (broken-in) and delivered with these greases. The bearings shall be cleaned before lubrication with military greases.

3.15.5.2 Oil lubrication. Unless otherwise preserved for packaging (see 5.1), all components shall be filled to the operating level with winter grades of military oils designated for use in the temperature range of 0° F to 32° F (minus 20° C to 0° C). Each lubricated component shall be tagged in a conspicuous place to indicate the temperature ranges and grades of lubricants used.

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3.16 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus and moisture growth by an overall treatment with a varnish conforming to MIL-V-173 with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.17 Identification marking. The drilling machine shall be identified in accordance with MIL-STD-130. Identification plates shall conform to MIL-P-514.

3.18 Instruction plates. Each drilling machine shall be equipped with instruction plates, including warnings and cautions, describing any special important procedures to be followed in operating and servicing the machine. Instruction plates shall conform to MIL-P-514.

3.19 Treatment and painting. The portions of the drilling machine and its components and parts normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A.

3.20 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished.

3.21 Items to be furnished with the drilling machine. No shipment of drilling machines shall be made unless repair parts, maintenance tools, operating tools, and accessories are included with the shipment or unless approval for shipment of the drilling machines without such items has been received from the contracting officer.

3.22 Toolbox. The drilling machine shall be provided with a weatherproof toolbox made of steel. When the toolbox is mounted on the drilling machine, the toolbox lid shall remain in the fully open position either by its own weight or by holding devices. The fasteners shall be attached to the toolbox by either bolts or rivets. A hasp which will accomodate a lock

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conforming to MS35647 shall be provided for locking the toolbox. The toolbox shall be fastened to the drilling machine in a protected position. A drain hole shall be provided in the bottom of the toolbox. The toolbox shall be of a size to hold the operating and maintenance tools and repair parts.

3.23 Workmanship. All parts, components, and assemblies of the drilling machine including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scales, flux, and other harmful extraneous material. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges and corners are required.

3.23.1 Steel fabrication. Steel used in the fabrication of the drilling machine shall provide original quality surface finish and shall be free from kinks and sharp bends. The forming of material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Burned surfaces of flame-cut material shall be free of slag. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

3.23.2 Welders and welding.

3.23.2.1 Welders. Before assigning any welder to manual or machine welding work covered by this specification, the contractor shall provide the contracting officer with active updated certification in accordance with either of the following listed codes for the type of welding operations to be performed:

AWS D1.1 - Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

3.23.2.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.

3.23.2.3 Aluminum welding. Aluminum welding shall be done by the metal-inert-gas method or any other method that will provide equivalent mechanical properties of the filler metal to base metal combinations as specified in Section 3. All aluminum welds shall be made by the automatic-machine method unless otherwise specified. All welds shall be made at a uniform rate of speed. Complete and uniform penetration and fusion of the metals shall be obtained on all welds. Aluminum welds may be ground, filed, wire brushed,

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or chipped but shall not be hammered. Preheating for welding is permissible provided the temperature does not exceed 400° F (204° C) for a total time of 30 minutes. No process requiring the use of welding flux shall be used.

3.23.3 Brazing. Brazing of steels, copper alloys, and nickel alloys shall conform to MIL-B-7883.

3.23.4 Castings. Castings shall be free from injurious defects. Surface shall be free from fused on sand, and shall be reasonably smooth. Runners, risers, fins and other cast on pieces shall be removed.

3.23.5 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Boltholes in structural members subject to variable loads shall be punched or drilled. Washers or lockwashers shall be provided where necessary. All bolts, nuts, and screws shall be correctly torqued.

3.23.6 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.23.7 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.6).
(e) Inspection of packaging (see 4.7).

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4.3 Preproduction inspection.

4.3.1 Examination. The drilling machine shall be examined as specified in 4.5.1. Presence of one or more major defects shall be cause for rejection.

4.3.2 Tests. The drilling machine shall be tested as specified in 4.5.2. Failure to pass any test shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.3), an initial production drilling machine(s) will be selected at random by the Government from the first five drilling machines being produced from production tooling and will be examined as specified in 4.5.1 and tested as specified in 4.5.2 through 4.5.2.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production drilling machine shall not exclude the remaining drilling machines from the quality conformance inspection and acceptance provisions specified in Section 4.

4.4.1 Inspection failure. Failure of an initial production drilling machine to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production drilling machine(s) and shall be cause for refusal by the Government to continue acceptance of production drilling machines until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on drilling machines previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all drilling machines accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each drilling machine shall be examined for characteristics as specified in Table I. AQL shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

4.5.2 Tests. Each drilling machine shall be tested as specified in 4.5.2.3. Failure to pass this test shall be cause for rejection.

4.5.2.1 Operational test. With the fuel tank full, the drilling machine shall be mounted on any rail ranging between 80 and 155 pounds (32 and 70 kg) per yard (914 mm). Not less than 25 holes shall be drilled with each

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size of bit furnished. One of the holes shall be drilled near a switch where the rails are 14-inches (356 mm) apart. Inability to respond to controls or nonconformance to 3.9, 3.10, or 3.12 shall constitute failure of this test.

4.5.2.2 Drilling speed test. With the drilling machine mounted on a 115-pound-per-yard rail, five holes shall be drilled through the web of the rail using the 1-1/8 inches (29 mm) drill specified in 3.11. Inability of the drilling machine to drill each hole through the web of the rail in 1-1/2 minutes shall constitute failure of this test.

4.5.2.3 Individual tests. Each drilling machine shall be operated for not less than 15 minutes and shall drill through the web of a 115-pounds-(52 kg)-per-yard (914 mm) rail using a 1-1/8 inch (29 mm) drill. Improper fit of any part or component, inability of components to respond to controls, or overheating of bearings and clutch shall constitute failure of this test.

Table I. Examination Schedule

Number	Characteristic	Requirement Paragraph
Major		
101.	Dimensions that affects form, fit, or function.	3.1
102.	Material not as specified.	3.4
103.	Safety not as specified.	3.6
104.	Overall dimensions and weight not as specified.	3.8
105.	Frame not as specified.	3.10
106.	Drill bits not as specified.	3.11
107.	Drill chuck not as specified.	3.12
108.	Engine not as specified.	3.13
109.	Power transmission not as specified.	3.14
110.	Clutch not as specified.	3.14.1

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Table I. Examination Schedule (Cont'd)

Number	Characteristic	Requirement Paragraph
111.	Hydrodynamic drive unit not as specified.	3.14.2
112.	Lubrication not as specified.	3.15
113.	Lubricants not as specified.	3.15.1
114.	Lubrication fittings not as specified.	3.15.2
115.	Pressure-release device not as specified.	3.15.3
116.	Enclosures with integral reservoir of lubricant not as specified.	3.15.4
117.	Grease lubrication not as specified.	3.15.5.1
118.	Oil lubrication not as specified.	3.15.5.2
119.	Fungus and moisture resistance not as specified.	3.16
120.	Instruction plates not as specified.	3.18
121.	Steel fabrication not as specified.	3.23.1
122.	Welders not as specified.	3.23.2.1
123.	Welding not as specified.	3.23.2.2
124.	Aluminum welding not as specified.	3.23.2.3
125.	Brazing not as specified.	3.23.3
126.	Castings not as specified.	3.23.4
127.	Bolted connections not as specified.	3.23.5
128.	Riveted connections not as specified.	3.23.6
129.	Machine work not as specified.	3.23.7

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Table I. Examination Schedule (Cont'd)

Number	Characteristic	Requirement Paragraph
Minor		
201.	Identification marking not as specified.	3.17
202.	Treatment and painting not as specified.	3.19
203.	Repair parts and maintenance tools not furnished as specified.	3.20
204.	Items not furnished with the drilling machine as specified.	3.21
205.	Toolbox not as specified.	3.22
206.	Workmanship not as specified.	3.23

Table II. Examination Schedule (Packaging)

Number	Characteristics	Requirement Paragraph
Major		
130.	The preservation of the drilling machines not Level A or Commercial as specified by MIL-M-18058.	5.1
131.	Packing not Level A, Level B, or Commercial as specified.	5.2
132.	Each complete drilling machine not packed for Level A requirements of MIL-M-18058.	5.2.1
133.	Each complete drilling machine not packed for Level B requirements of MIL-M-18058.	5.2.2
134.	Each complete drilling machine not packed for Commercial as specified.	
Minor		
207.	Marking not in accordance with MIL-STD-129.	5.3

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4.6 Inspection comparison. When initial production inspection is specified, the Government may select drilling machines at any time during the contract production period and subject these drilling machines to the examination and tests specified in 4.4 to determine that the quality of the selected drilling machine is equal to the quality standards established during initial production inspection. The inspection will be performed by the Government at a site selected by the Government. Drilling machines will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model(s) and initial production drilling machines.

4.6.1 Inspection failure. Failure of any drilling machine, when inspected as specified in 4.7, may be considered cause for refusal by the Government to continue acceptance of drilling machines until objective evidence furnished by the contractor reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.7 Inspection of packaging.

4.7.1 Preproduction pack inspection.

4.7.1.1 Examination. The preproduction pack shall be examined for the characteristics listed in Table II. Presence of one or more defects shall be cause for rejection.

4.7.2 Quality conformance inspection of pack.

4.7.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.2.2 Examination. Each drilling machine shall be examined for the characteristics listed in Table II. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

5. PACKAGING

5.1 Preservation, packing, and marking. Drilling machines shall be preserved, packed, and marked in accordance with MIL-M-18058. The degree of preservation shall be Level A or Commercial and the degree of packing shall be Level A, Level B, or Commercial as specified (see 6.2). When Level A preservation is specified, the engine and engine accessories shall be preserved in accordance with MIL-E-10062 for Type II preparation.

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6. NOTES

6.1 Intended use. The drilling machine is intended for use in railway track construction and maintenance-of-way work.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2.1).
(c) When initial production inspection is required and number of drilling machines to be furnished, when applicable (see 3.3).
(d) Whether gasoline or diesel engine is required (see 3.13).
(e) When a hydrodynamic drive is required (see 3.14.2).
(f) Applicable list of repair parts and maintenance tools (see 3.20).
(g) When aluminum welding other than specified is required (see 3.23.2.3).
(h) When brazing other than specified is required (see 3.23.3).
(i) Degree of preservation and degree of packing required (see 5.1).

6.3 Lubricants. When requested by the contract, the contracting officer should obtain a list of military lubricants applicable to the drilling machine covered by this specification (see 3.15.1).

6.4 Preproduction model. Any changes or deviations of production drilling machines from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish drilling machines conforming to this specification.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 2230-A036

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-D-3358C(ME) Drilling Machines, Rail, Gasoline or Diesel, Portable

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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